1. **Simple Interest** Suppose Liz borrows \$1000 and agrees to pay it back in 3 years with 5% simple interest added each year.

   (a) How much **interest** will she pay? In addition to your answer, write down your computation, including the decimal values you would plug into a calculator/spreadsheet.

   (b) How much will she owe **in total** at the end of three years? Write down the computation and the answer.

2. **Annual Interest Rate (APR)** Annual interest rates are pro-rated by time.

   (a) Definition: An APR of $r\%$ paid n times per year means

   (b) So an APR of 5% paid **quarterly** means

3. Determine n for each word below.

word:	annually	semi-annually	quarterly	monthly	weekly	daily
n times per year						

Compound Interest

4. What is **compound** interest and how is it different from **Simple** Interest?

5. Suppose \$1000 will be invested in an account that accrues **compound** interest at an APR of 3% compounded **monthly**. We will set up a spreadsheet to model this account assuming no other deposits or withdrawals.

   (a) Open a new spreadsheet and type column headers in cells `A1, B1, C1, D1, E1, F1`. (See below.)

   (b) In row 2, type the variables for reference. (hint: `A2` is r.)

   (c) In row 3, enter the **start** values for this specific problem. Assume time t is in years.

   (d) In rows 4 and 5, enter the values/formulas and the end of 1 month and 2 months.

   (e) Use the pull down feature to determine the values in the account at the end of 2 years.

	A	B	C	D	E	F
1	`Interest Rate`	`Times per Year`	`Principal`	`Time Elapsed`	`Interest`	`Total`
2	`r`					
3						
4						
5						

6. Write out the month-by-month computations and look for a pattern.

7. **Compound Interest Formula:** A principal value of P dollars earning r% interest compounded n times per year for t years will result in a total of A dollars where

8. **Effective Annual Interest Rate:**

9. What is the effective annual interest rate on \$1000 invested at a annual percentage rate of 3% compounded monthly?